Hackathon Problem Statements

Regd No: 25091A3242, 25091A3250

Problem Statement No: 13

Farm Equipment Sharing/Rental Matching System Create a peer-to-peer farm equipment rental platform that uses bipartite matching algorithms to pair equipment owners with nearby farmers needing tractors, harvesters, or sprayers during peak seasons. Apply clustering to identify demand hotspots by season and geography for better equipment placement. Use OOP to model equipment, owners, and renters with booking and pricing logic. This reduces capital burden on small farmers who cannot afford individual machinery.

Regd No: 25091A3275, 25091A32A0

Problem Statement No: 64

Library Book Recommendation System Build a book recommendation engine using collaborative filtering basics (user-based or item-based similarity) on library borrowing history datasets to suggest relevant books to students. Use OOP to model books, students, and borrowing records. This can increase library engagement and help students discover relevant academic and extracurricular reading material more effectively than static catalog browsing.

Regd No: 25091A32J8, 25091A32K6

Problem Statement No: 76

Plastic Waste Segregation Efficiency Tracker Design a tracking system that uses classification algorithms on waste composition audit data to measure segregation efficiency (wet/dry/recyclable) across different city wards, identifying areas needing awareness campaigns. Use OOP to model wards, waste categories, and audit records. This supports municipal solid waste management compliance with segregation-at-source regulations.

Regd No: 25091A32K0, 25091A32Q1

Problem Statement No: 83

Biodiversity Index Calculator Build a biodiversity assessment tool using statistical aggregation methods on species-count survey datasets from a given region to compute standard biodiversity indices (like Shannon diversity index) and track ecological health over time. Use OOP to model species, survey records, and regional biodiversity profiles. This supports ecological researchers and forest departments in monitoring conservation program effectiveness.

Regd No: 25091A32G4, 25091A32D9

Problem Statement No: 130

Public Toilet/Sanitation Facility Mapping & Quality Scorer Design a sanitation facility mapping tool using classification algorithms on facility condition survey data (cleanliness, water availability, functionality) to generate quality scores and identify facilities needing urgent maintenance. Use OOP to model facilities, inspections, and quality metrics. This supports Swachh Bharat-aligned sanitation infrastructure monitoring at the city or district level.

Regd No: 25091A3262, 25091A3204

Problem Statement No: 147

Secure File-Sharing System with Access Control Build a secure file-sharing system using an OOP-based role hierarchy (admin, editor, viewer) to enforce granular access control permissions, combined with basic encryption for stored files. Use OOP inheritance and interfaces to model different user roles and permission sets cleanly. This demonstrates practical secure-by-design principles applicable to any organization handling sensitive documents.